10C 1 ALCOHOLS AND SOME OF THEIR REACTIONS

SPECIFICATION REFERENCE
10.15 10.16 10.17 CP6

LEARNING OBJECTIVES

- Understand the nomenclature of alcohols and be able to draw their structural, displayed and skeletal formulae.
- Understand the distinction between primary, secondary and tertiary alcohols.
- Understand the combustion, halogenation and dehydration reactions of alcohols.



▲ **fig A** Sanitising hand gels containing alcohols are used in hospitals to reduce infection risks to patients.

WHAT ARE ALCOHOLS?

The alcohols are a homologous series of compounds with the general formula $C_nH_{2n+1}OH$. Think of them as the result of replacing a hydrogen atom in a hydrocarbon with a hydroxyl group. You have already met alcohols in **Topic 10B.1** as the products of the hydrolysis of halogenoalkanes.

The symbol R can be used to represent an alkyl group, as with halogenoalkanes. The formula of an alcohol can be simplified to ROH.

NAMING ALCOHOLS

We have already discussed how to name alcohols in **Topic 4A.4**, but **table A** shows some examples to remind you.

STRUCTURE	NAME
$CH_3–CH(OH)–CH_3$	propan-2-ol
$CH_3–CH_2–CH_2–CH_2OH$	butan-1-ol
$(CH_3)_3C–CH_2OH$	2,2-dimethylpropan-1-ol
$CH_2(OH)–CH(OH)–CH_2OH$	propane-1,2,3-triol

table A Examples of how to name alcohols.

You will have come across the last example in foods, medicines and personal care products. Its common names are glycerol and glycerine. You can see how the IUPAC system can be adapted to name compounds with more than one OH group.

CLASSIFYING ALCOHOLS

You have met the terms primary, secondary and tertiary applied to carbocations (in **Topic 5A.4**) and to halogenoalkanes (in **Topic 10B.1**).

Alcohols are classified in a similar way, depending on the number of alkyl groups joined to the C atom bonded to the hydroxyl group. **Table B** shows some examples.

STRUCTURE	ABBREVIATED FORMULA	NUMBER OF ALKYL GROUPS	CLASSIFICATION
$CH_3–CH_2–CH_2OH$	ROH	1	primary
$CH_3–CH(OH)–CH_3$	R_2CHOH	2	secondary
$(CH_3)_2C(OH)–CH_2–CH_3$	R_3COH	3	tertiary

table B Examples of how to classify alcohols.

REACTIONS

In this topic, the reactions of alcohols we cover are:

- combustion
- conversions to halogenoalkanes
- dehydration to alkenes.

Reaction mechanisms are not required for any of these reactions.

COMBUSTION

You have already seen in **Topic 4B.3** that alcohols are used as biofuels. If combustion is complete, the products are carbon dioxide and water. This is the equation for the complete combustion of ethanol:

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

CONVERSIONS TO HALOGENOALKANES

These reactions involve replacing the hydroxyl group in an alcohol molecule with a halogen atom. The reaction is known as **halogenation**. However, just adding a halogen to an alcohol does not work. A different method is needed for each halogen.

Chlorination is carried out using phosphorus(V) chloride (a white solid, also known as phosphorus pentachloride). The reaction is very vigorous at room temperature, so the alcohol and phosphorus(V) chloride reaction mixture does not need heating. There are also two inorganic products: phosphoryl chloride and hydrogen chloride. This is the equation for the reaction with propan-1-ol:

$$CH_3CH_2CH_2OH + PCl_5 \rightarrow CH_3CH_2CH_2Cl + POCl_3 + HCl$$

Chlorination of tertiary alcohols can be done in a different way, using a method that does not work well for primary and secondary alcohols. The alcohol needs only to be mixed (by shaking) with concentrated hydrochloric acid at room temperature. This is the equation for the reaction with 2-methylpropan-2-ol:

$$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$$

Bromination is carried out using a mixture of potassium bromide and about 50% concentrated sulfuric acid. The reaction mixture is warmed with the alcohol. It is better to write two equations, rather than one, as the inorganic reagents first react together to form hydrogen bromide.

The other inorganic product is either potassium hydrogensulfate or potassium sulfate.

$$KBr + H_2SO_4 \rightarrow KHSO_4 + HBr$$

or

$$2KBr + H_2SO_4 \rightarrow K_2SO_4 + 2HBr$$

This is the equation for the reaction with butan-1-ol:

$$CH_3CH_2CH_2CH_2OH + HBr \rightarrow CH_3CH_2CH_2CH_2Br + H_2O$$

EXAM HINT

The sulfuric acid must be no more concentrated than a 50% solution. More concentrated sulfuric acid would oxidise bromide ions to bromine and so result in different products.

Iodination is carried out using a mixture of red phosphorus and iodine. The reaction mixture, including the alcohol, is heated under reflux. As with bromination, it is better to write two equations, as the inorganic reagents first react to form phosphorus(III) iodide:

$$2P + 3I_2 \rightarrow 2PI_3$$

This is the equation for the reaction with ethanol:

$$3C_2H_5OH + PI_3 \rightarrow 3C_2H_5I + H_3PO_3$$

The inorganic product is phosphonic acid (often known as phosphorous acid).

DEHYDRATION TO ALKENES

Dehydration is done by heating the alcohol with concentrated phosphoric acid. The reaction is similar to the elimination reaction of a halogenoalkane, with the OH group and a hydrogen atom from an adjacent carbon atom being removed and a C=C double bond formed in the carbon chain.

You can see why the reaction is described as dehydration, as water is the only inorganic product. These are the equations for a reaction in which there are two possible products, starting with butan-2-ol:

$$CH_3CH(OH)CH_2CH_3 \rightarrow CH_2{=}CHCH_2CH_3 + H_2O$$

but-1-ene

and

$$CH_3CH(OH)CH_2CH_3 \rightarrow CH_3CH{=}CHCH_3 + H_2O$$

but-2-ene

Remember that but-2-ene actually exists as a pair of *E–Z* (*cis-trans*) isomers. It is therefore more correct to state that there are three products – but-1-ene and the two isomeric forms of but-2-ene.

The formula for phosphoric acid does not appear in the equation. This is because the water formed in the reaction mixes with the concentrated phosphoric acid to dilute the acid.

LEARNING TIP

When you balance an equation for the combustion of an alcohol, remember that there is already one atom of oxygen in an alcohol molecule.

CHECKPOINT

SKILLS REASONING

1. Write an equation for the complete combustion of butanol, C_4H_9OH.
2. When butan-2-ol is dehydrated, there is more than one organic product. Explain why there is only one pair of *E/Z* isomers when pentan-3-ol is dehydrated.

SUBJECT VOCABULARY

halogenation a reaction where the hydroxyl group in an alcohol molecule is replaced by a halogen atom

dehydration a reaction where the hydroxyl group from an alcohol molecule, and a hydrogen atom from an adjacent carbon atom, are removed, forming a C=C double bond

10C 2 OXIDATION REACTIONS OF ALCOHOLS

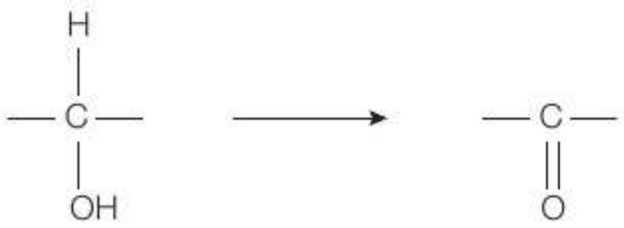
SPECIFICATION REFERENCE
10.18 10.19(i) CP7

LEARNING OBJECTIVES

- Understand the oxidation reactions of primary and secondary alcohols by potassium dichromate(VI) to form aldehydes, carboxylic acids and ketones.
- Understand the techniques of distillation and heating under reflux used to maximise the yields of different products.

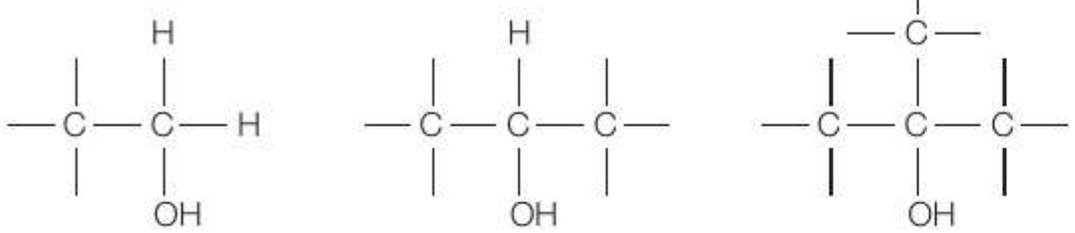
BACKGROUND

There are different ways of considering oxidation and reduction. In this topic, the best way to consider oxidation is as the loss of hydrogen from an alcohol molecule. Unlike dehydration, covered in **Topic 10C.1**, oxidation affects only one carbon atom. The atoms removed from an alcohol molecule are the hydrogen of the OH group and a hydrogen atom from the carbon atom joined to the OH group, as shown below.



The organic product contains a C=O group, known as a carbonyl group.

The diagrams below should help you to understand why primary and secondary alcohols, but not tertiary alcohols, can be oxidised in this way.



Only the primary and secondary structures have a hydrogen atom on the C of the C—OH group – the tertiary structure does not.

THE PRODUCTS OF OXIDATION

KETONES

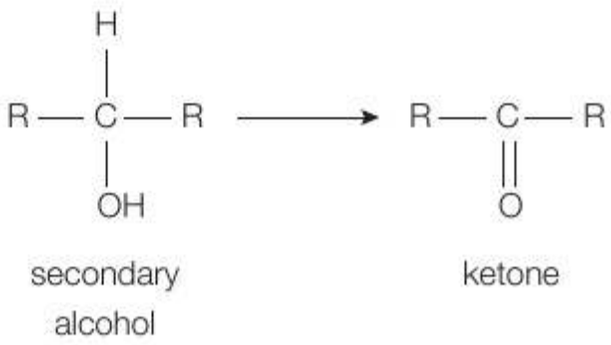
When a secondary alcohol is oxidised, the organic product belongs to a homologous series called **ketones**. A simplified formula for a ketone is RCOR. Using the same symbol R twice suggests that the two alkyl groups are the same (for example, two ethyl groups). This is not necessarily the case, but if the two alkyl groups are meant to be different, then the R symbols can be altered slightly. For example, one could be R and the other could be R'. Another way to show that the alkyl groups are different is for one to be R_1 and the other R_2.

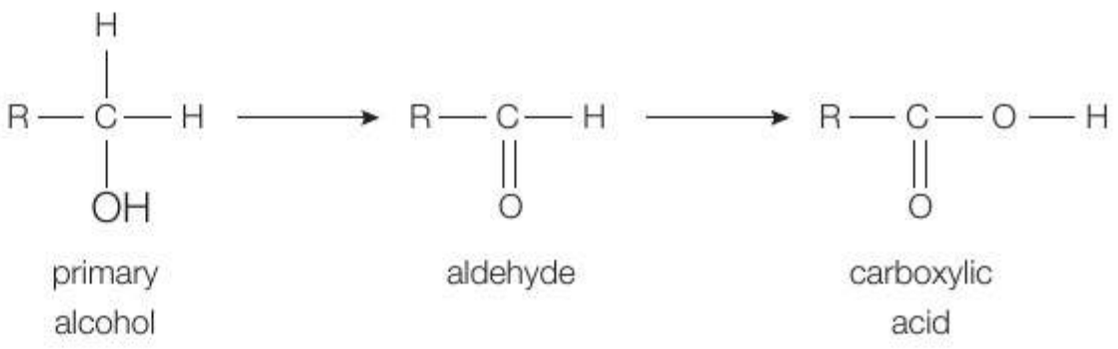
ALDEHYDES AND CARBOXYLIC ACIDS

When a primary alcohol is oxidised, the organic product belongs to a homologous series called **aldehydes**. A simplified formula for an aldehyde is RCHO. This should *not* be written as RCOH, which would imply that the molecule contains an OH group.

There is a complication with aldehydes that is not the case with ketones. This is that aldehydes are more easily oxidised than alcohols, so when a primary alcohol is oxidised, the aldehyde formed may be oxidised further. When an aldehyde is oxidised, the process involves the gain of an oxygen atom, not the loss of hydrogen. The oxygen atom gained goes between the C and H of the CHO group. The organic product belongs to a homologous series called **carboxylic acids**. A simplified formula for a carboxylic acid is RCOOH.

These reactions are summarised below.





The usual reagent for these oxidation reactions is a mixture of potassium dichromate(VI) and dilute sulfuric acid. Unlike with the inorganic reagents used previously, equations involving this mixture do not need to be written (because they are very complicated). Instead, the oxidising agent is represented by [O]. This symbol simply represents an oxygen atom provided by the oxidising agent. Whenever the mixture of potassium dichromate(VI) and sulfuric acid is used as an oxidising agent, there is a colour change from orange to green.

So now we can write some equations for examples of oxidation reactions. The names of the organic products are given, but a fuller study of these products is outside the scope of IAS.

Propan-1-ol to propanal:

$$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$$

Propanal to propanoic acid:

$$CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$$

Propan-2-ol to propanone:

$$CH_3CH(OH)CH_3 + [O] \rightarrow CH_3COCH_3 + H_2O$$

You do not need to know the mechanisms of any of these reactions.

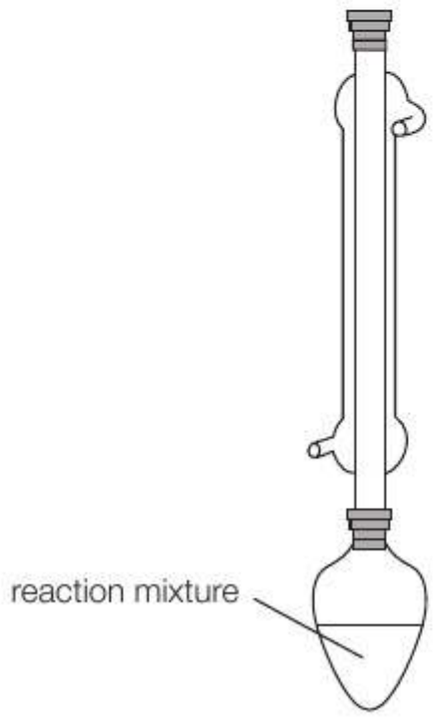
DIFFERENT PRACTICAL TECHNIQUES

Because of the easier oxidation of aldehydes compared with alcohols, two different techniques are used. These are:

- heating under reflux
- distillation with addition.

HEATING UNDER REFLUX

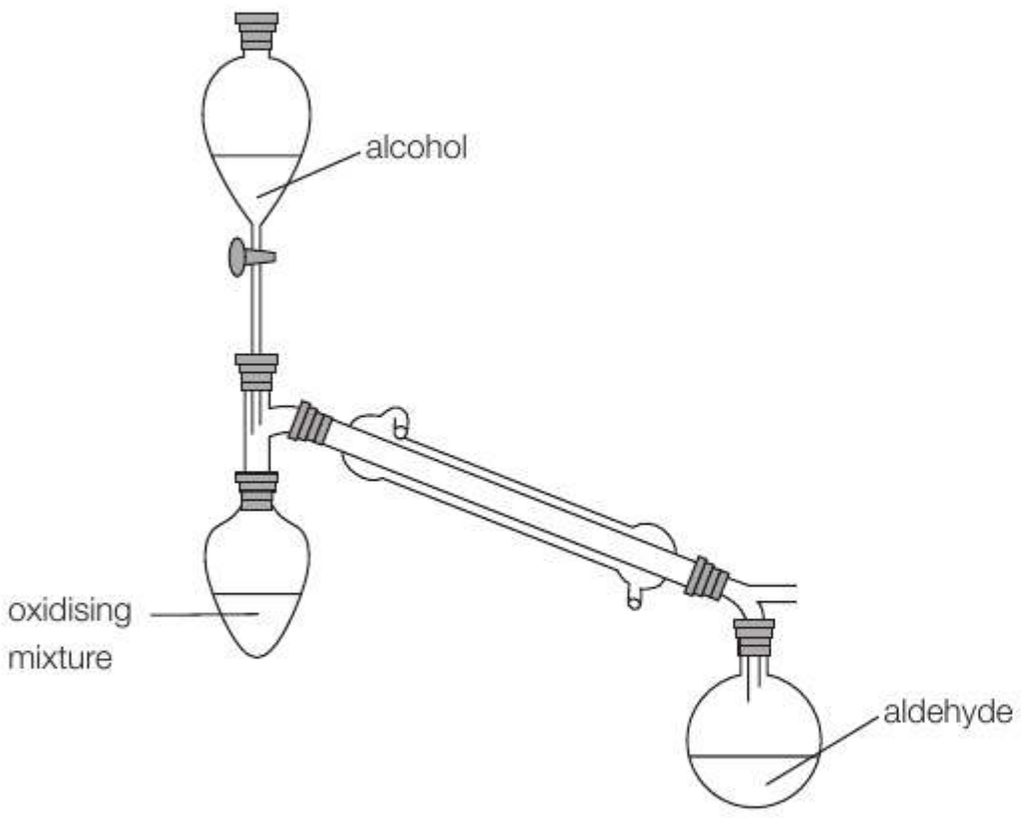
When the oxidation is intended to be complete (to obtain a ketone or a carboxylic acid), the technique used is **heating under reflux**. You can see the apparatus for this technique in the diagram.



In this apparatus, the products of oxidation stay in the reaction mixture, because if they do boil off, they condense in the vertical condenser and return to the heating flask.

DISTILLATION WITH ADDITION

When the oxidation is intended to be incomplete (to obtain an aldehyde, and not a carboxylic acid), the technique used is **distillation with addition**. The apparatus for this technique is shown in the diagram.



In this apparatus, only the oxidising agent is heated, and the alcohol is slowly added to the oxidising agent. When the aldehyde is formed, it immediately distils off (it has a much lower boiling temperature than the alcohol used to make it), and is collected in the receiver.

LEARNING TIP

Practise writing equations to show how a given alcohol can be separately dehydrated and oxidised. This will help you to understand the difference between the two reactions.

CHECKPOINT

1. Butan-1-ol can be dehydrated to form one organic product, but can be oxidised to form two organic products. Give the names and structures of these three products.
2. Pentan-2-ol can be dehydrated to form more than one organic product, but can be oxidised to form only one organic product. Give the names and structures of these products.
3. There are four isomeric alcohols with the formula C_4H_9OH. Draw a structural formula for each isomer and use the formulae to explain which ones can be oxidised by heating with acidified potassium dichromate(VI).

SUBJECT VOCABULARY

ketone one of a homologous series of organic compounds formed by oxidation of secondary alcohols, formula RCOR

aldehyde one of a homologous series of organic compounds formed by the partial oxidation of primary alcohols, formula RCHO

carboxylic acid one of a homologous series of organic compounds formed by the complete oxidation of primary alcohols, formula RCOOH

heating under reflux heating a reaction mixture with a condenser fitted vertically

distillation with addition heating a reaction mixture, but adding another liquid and distilling off the product as it forms

10C 3 PURIFYING AN ORGANIC LIQUID

SPECIFICATION REFERENCE
10.19(ii-v)

LEARNING OBJECTIVES

- Know when and how to use the following techniques in the preparation and purification of a liquid organic compound: simple distillation, fractional distillation, solvent extraction, drying, and boiling temperature determination.

BACKGROUND

So far, you have met many different reactions that can be used in the preparation of organic compounds. In most cases the intended organic product is not pure. It could be contaminated with:

- unreacted starting materials
- other organic products
- the inorganic reagents used, or the inorganic products formed from them
- water.

This means that organic chemists need to use several techniques to separate the intended product from a reaction mixture. These techniques will be different, depending on whether the intended product is a gas, a liquid or a solid. In this topic, we consider only the techniques used to purify an organic liquid.

The techniques we consider are:

- simple distillation
- fractional distillation
- solvent extraction
- drying
- boiling point determination.

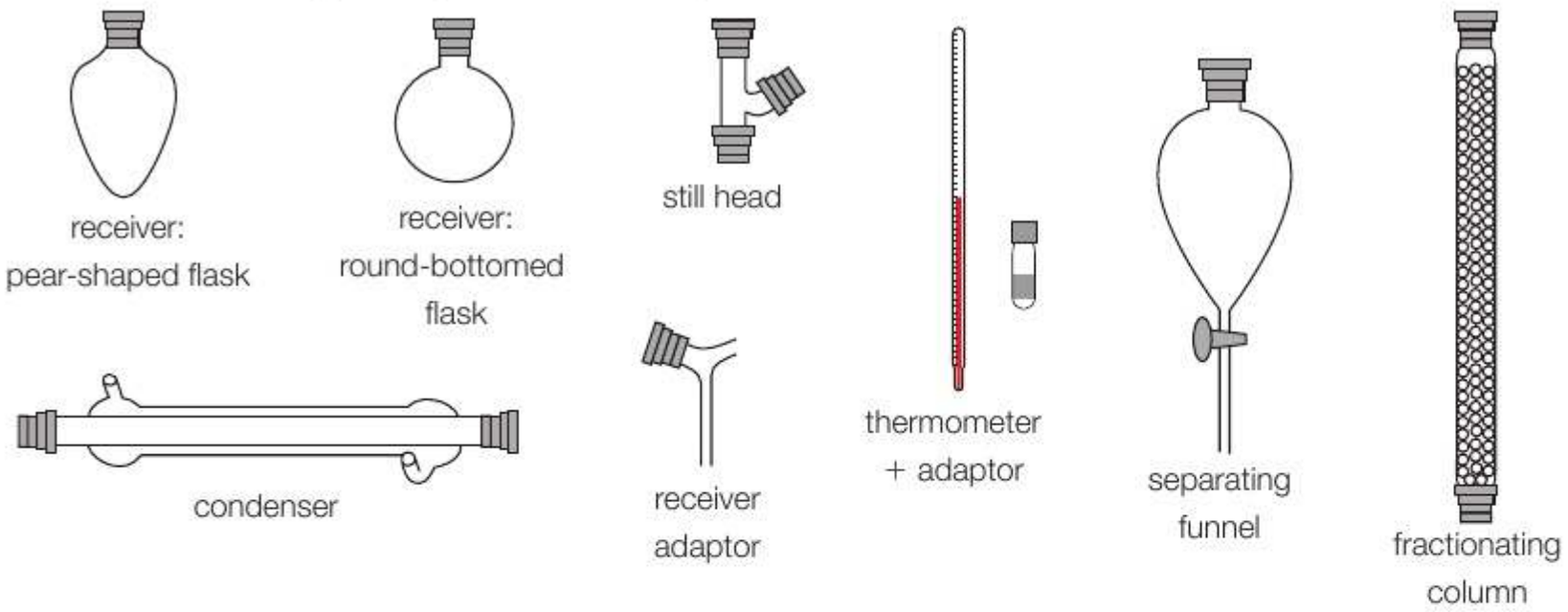
APPARATUS

In a chemistry textbook, you will see many different diagrams of apparatus used in organic chemistry. You will have some of these pieces of apparatus in your laboratory, but you will probably not be able to set up every piece of apparatus you see in the book.

In simple laboratory experiments not involving organic compounds, you are likely to use pieces of glassware connected by tubing and corks or bungs. This kind of apparatus is relatively inexpensive to buy and easy to use. If one of the gases you are using, such as carbon dioxide, leaks from the apparatus, this is not a major problem.

Organic compounds are more of a problem because they may be flammable, or toxic, or both. They may also attack corks and bungs, and so increase the risk of leaks and contamination. One solution to these problems is to use a type of apparatus made only (or mostly) of glass, and which can be fitted together tightly using ground-glass joints.

Fig A shows a selection of this type of apparatus. Using these eight pieces of apparatus, you can create a wide variety of experimental set-ups.



▲ **fig A** A selection of apparatus made mainly of glass, ideal for experiments using organic compounds.



▲ **fig B** This is just one of the many apparatus set-ups using ground-glass joints.

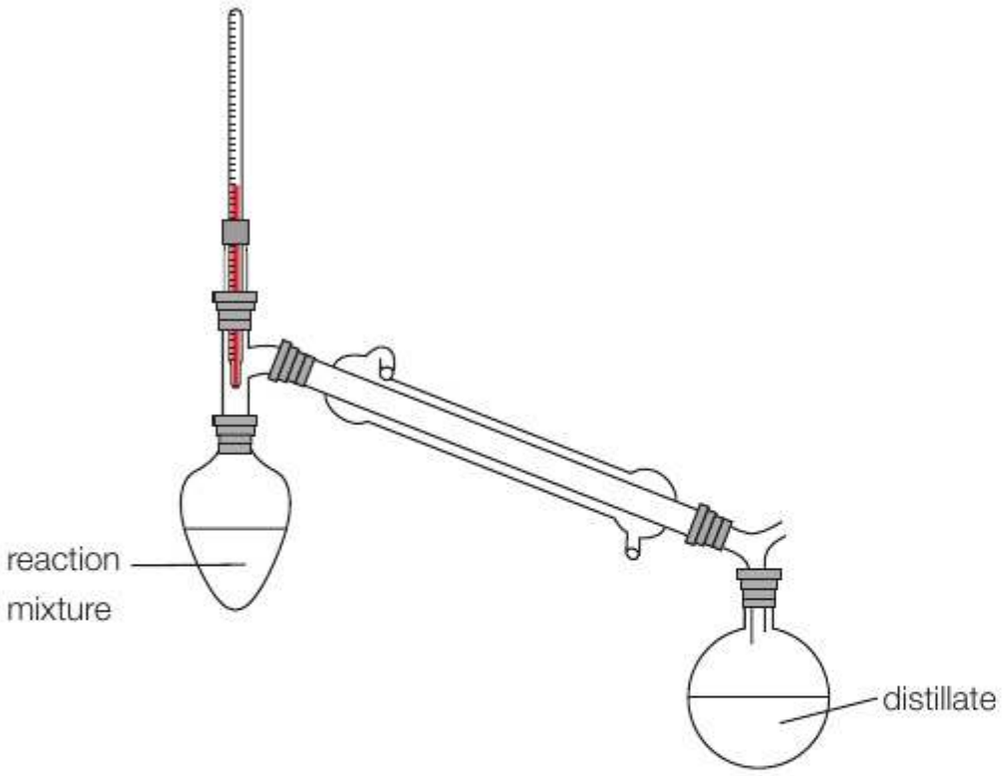
Heating is needed in many experiments. You can do this using a Bunsen burner, but often an electric heating mantle or a hot water bath or hot oil bath is preferred for safety reasons.

SIMPLE DISTILLATION

Here is a summary of the process of **simple distillation**.

- Distillation of an impure liquid involves heating the liquid in a flask connected to a condenser.
- The liquid with the lowest boiling temperature evaporates or boils off first and passes into the condenser first. This means it can be collected in the receiver separately from any other liquid that evaporates later.
- The purpose of the thermometer is to monitor the temperature of the vapour as it passes into the condenser. If the temperature remains steady, this is an indication that one compound is distilling over. If, after a while, the temperature begins to rise, this indicates that a different compound is beginning to distil over.

Fig C shows the apparatus used for simple distillation.

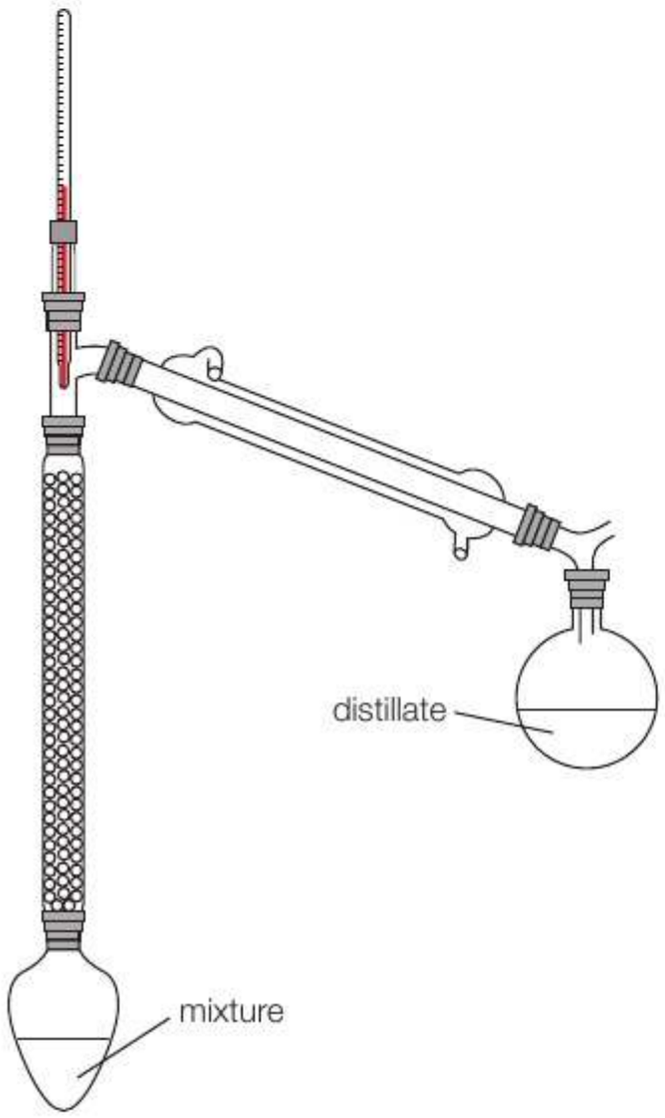


▲ **fig C** Apparatus used for simple distillation.

FRACTIONAL DISTILLATION

Here is a summary of the process of **fractional distillation**.

- Fractional distillation uses the same apparatus as simple distillation, but with a fractionating column between the heating flask and the still head (**fig D**).
- The column is usually filled with glass beads or pieces of broken glass, which act as surfaces on which the vapour leaving the column can condense. It can then be evaporated again as more hot vapour passes up the column.
- Effectively, the vapour undergoes several repeated distillations as it passes up the column. This provides a better separation.



▲ **fig D** Apparatus used for fractional distillation.

Fractional distillation takes longer than simple distillation, and is best used when the difference in boiling temperatures is small, and when there are several compounds to be separated from a mixture.

ADVANTAGES AND DISADVANTAGES

The advantages of using simple distillation, rather than fractional distillation, are that it is easier to set up and it is quicker.

The disadvantage is that it does not separate the liquids as well as fractional distillation. Simple distillation should only be used if the boiling temperature of the liquid being purified is very different from the other liquids in the mixture, ideally a difference of more than 25 °C.

SOLVENT EXTRACTION

As the name suggests, this method involves using a solvent to remove the desired organic product from the other substances in the reaction mixture. There are several solvents that can be used, but the choice depends mainly on these features.

- The solvent added should be immiscible (i.e. does not mix) with the solvent containing the desired organic product.
- The desired organic product should be much more soluble in the solvent added than in the reaction mixture.

Here is a summary of the process of **solvent extraction**.

- Place the reaction mixture in a separating funnel, and then add the chosen solvent – it should form a separate layer.
- Place the stopper in the neck of the funnel and gently shake the contents of the funnel for a while.
- Allow the contents to settle into two layers.
- Remove the stopper and open the tap to allow the lower layer to drain into a flask. Then pour the upper layer into a separate flask.



▲ **fig E** Apparatus used for solvent extraction.

If a suitable solvent is used and the method is followed correctly, most of the desired organic product will have moved into the added solvent. It is better to use the solvent in small portions than in a single large volume (for example, four portions of 25 cm^3 rather than one portion of 100 cm^3) because this is more efficient. Using more portions of solvent, but with the same total volume, removes more of the desired organic product.

The desired organic product has been removed from the reaction mixture, but is now mixed with the added solvent. Simple distillation or fractional distillation now has to be used to separate the desired organic product from the solvent used.

DRYING

Many organic liquids are prepared using inorganic reagents, which are often used in aqueous solution. A liquid organic product may partially or even completely dissolve in water, so water may be an impurity that needs to be removed by a drying agent. One important feature of a drying agent is that it does not react with the organic liquid.

There are several drying agents available, but the most common ones are anhydrous metal salts, often calcium sulfate, magnesium sulfate and sodium sulfate. What these compounds have in common is that they form hydrated salts, so when they come into contact with water in an organic liquid they absorb the water as water of crystallisation. Anhydrous calcium chloride can also be used for some organic compounds, although it does react with others and is soluble in alcohols.

Here is a summary of how to dry an organic liquid.

- The drying agent is added to the organic liquid and the mixture is swirled or shaken, and then left for a period of time.
- Before use, a drying agent is powdery, but after absorbing water it looks more crystalline.

- If a bit more drying agent is added, and it remains powdery, this is an indication that the liquid is dry.
- The drying agent is removed either by decantation (pouring the organic liquid off the solid drying agent), or by filtration.

TESTING FOR PURITY

Using one or more of the purifying techniques, you have purified an organic compound. But is it really pure?

For liquids, there is a simple way to test whether it is pure – measure its boiling temperature. Impurities raise the boiling temperature.

The boiling temperatures of pure organic compounds have been carefully measured and are widely available in data books and online. If you measure the boiling temperature of your organic compound, you can compare it with an accurate value and then make your decision about how pure it is.

The apparatus used depends on the volume of liquid available, and whether it is toxic or flammable. The apparatus used for simple distillation can be used.

A word of caution – this test may not be conclusive, because you may not be able to measure the boiling temperature of your organic compound accurately enough. Your thermometer might read too low or too high, so even if your measured boiling temperature exactly matches the one in the data book or online, you may wrongly assume that your compound is pure.

Also remember that different organic compounds can, by coincidence, have the same boiling temperature. For example, both 1-chloropentane and 2-methylpropan-1-ol boil at 108 °C.

LEARNING TIP

Give examples of when you would use fractional distillation instead of simple distillation. Think back to the advantages and disadvantages of both methods listed above.

CHECKPOINT

1. Describe the apparatus you could use to perform:
   (a) a simple distillation
   (b) fractional distillation
   (c) solvent extraction.
2. What is the purpose of using a thermometer during distillation?
3. What are the limitations of using the measurement of a boiling temperature as a way of assessing the purity of an organic liquid?

SUBJECT VOCABULARY

simple distillation a method used to separate liquids with very different boiling temperatures, by boiling and condensing

fractional distillation a method used to separate liquids with similar boiling temperatures, by boiling and condensing, using a fractionating column

solvent extraction a method used to separate a liquid from a mixture by causing it to move from the mixture to the solvent